

Chapter 10 In-Class Exercise: Building a Phone-Book Search

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1 Step 1: Organize the Dictionary

We will store each phone number as a dictionary key. The value will be a list containing the person's first and last name.

```
phone_book = {
    "562-555-0142": ["Maria", "Garcia"],
    "562-555-0188": ["Daniel", "Kim"],
    "562-555-0125": ["Elena", "Garcia"],
    "562-555-0171": ["Maria", "Lopez"],
    "562-555-0194": ["Aisha", "Patel"]
}
```

For one entry,

```
phone = "562-555-0142"
person = phone_book[phone]
```

Check the following for one entry: `person[0]` and `person[1]`.

2 Step 2: Add a Contact

Write a function that adds a new entry:

```
def add_contact(ph_book, phone, first, last):
    ph_book[phone] = [first, last]
```

Test it with

```
add_contact(
    ph_book,
    "562-555-0110",
    Marcus, "Johnson"
)
```

Print the contents of `ph_book` to check it.

3 Step 3: Search by First Name

Write a function that returns every phone number belonging to a given first name:

```
def search_first(phone_book, first):
    ...
    return matches
```

Test your function with a search.

4 Step 4: Search by Last Name

Complete the following function:

```
def search_last(phone_book, last):
    ...
    return matches
```

Test your function with a search.

5 Step 5: Search a Full Name

Write a third search function:

```
def search_name(phone_book, first, last):
    ...
    return matches
```

Test your function with a search.

6 Step 6: Put It Together

Write a short main program:

```
print("Phone Book Search")
print("1 - First name")
print("2 - Last name")
print("3 - Full name")
```

```
choice = input("Choose a search: ")
```

Use an `if/elif` statement to

- ask for the appropriate name information,
- call the correct search function, and
- store the returned list in `matches`.

Finally,

```
print("Phone numbers:")
```

```
for phone in matches:
    print(phone)
```

7 Challenge

If

```
len(matches) == 0
```

print

No matching contacts found.

Otherwise, print the number of matches found.